Journal of the Ceramic Society of Japan 115 [7] 425-428 (2007)

Paper

Synthesis of Calcium Phosphates Containing Metal Ions and Evaluation of their Catalytic Activity for the Decomposition of Hydrogen Peroxide

Masanobu KAMITAKAHARA, Akio TAKAHASHI,* Masao TANIHARA,*
Giichiro KAWACHI** and Chikara OHTSUKI**

Graduate School of Environmental Studies, Tohoku University, 6-6-20, Aoba, Aramaki, Aoba-ku, Sendai-shi 980-8579
**Graduate School of Materials Science, Nara Institute of Science and Technology, 8916-5, Takayama, Ikoma-shi, Nara 630-0192*
***Graduate School of Engineering, Nagoya University, Furo, Chikusa-ku, Nagoya-shi 464-8603*

Calcium phosphates with metal ions are expected to show interesting characteristics, such as catalytic activity, as well as high ability to adsorb organic substances. We assumed that the organic substances adsorbed on calcium phosphates could be decomposed by treatment with hydrogen peroxide (H_2O_2) if the catalytic activity for the decomposition of H_2O_2 can be introduced to calcium phosphates. We synthesized hydroxyapatite (HAp) containing Ni or Co ions, and octacalcium phosphate (OCP) containing Ni, Co or Mn ions. The HAp containing Co ions and the OCP containing Co or Mn ions showed catalytic activity for the decomposition of H_2O_2.

[Received April 17, 2007; Accepted June 14, 2007]

Key-words : *Calcium phosphate, Hydroxyapatite, Octacalcium phosphate, Metal ions, Catalyst*

1. Introduction

Hydroxyapatite (HAp, $Ca_{10}(PO_4)_6(OH)_2$) is one of the calcium phosphates and an inorganic component of bone. HAp is used as a bone substitute because of its high compatibility to body,[1] and as an adsorbent because of its specific adsorption properties against organic substances.[2] Utilizing the adsorption properties, HAp is expected to be a useful material that cleans the environment. However, HAp itself does not have the ability to decompose the organic substances adsorbed on it. A technique that can decompose the adsorbed substances is required to be able to reuse the HAp as an adsorbent. A number of attempts have been reported in which UV irradiation was used for TiO_2–HAp composites,[3),4)] HAp modified with Ti(IV)[5] and HAp itself.[6] However, in these systems, UV irradiation is essential and the decomposition of the adsorbed substances takes some time. We assumed that if the catalytic activity for the decomposition of hydrogen peroxide (H_2O_2) could be introduced to HAp, the adsorbed organic substances could be easily decomposed by a treatment with H_2O_2. As an adsorbent, we also focused on octacalcium phosphate (OCP, $Ca_8(HPO_4)_2(PO_4)_4 \cdot 5H_2O$), which has a similar structure to HAp and a layered structure.[7] Ohta *et al.*[8] reported that OCP showed adsorption behavior against proteins similar to HAp when its adsorption characteristic was investigated by liquid chromatography.

In this study, we attempted to prepare HAp and OCP containing metal ions, such as Ni, Co and Mn ions to introduce the catalytic activity on HAp and OCP for the decomposition of H_2O_2. It has been reported that the zeolite-encapsulated Ni and Co complexes can decompose H_2O_2,[9] and MnO_2 is known as famous catalyst for the decomposition of H_2O_2. The resultant samples were characterized, and their catalytic activity for the decomposition of H_2O_2 was examined.

2. Materials and methods

2.1 Synthesis of HAp containing metal ions

Synthesis of HAp containing Ni or Co ions was attempted. Calcium carbonate (Nacalai Tesque Inc., Japan) was heated up to 1373 K at a rate of 5 $K \cdot min^{-1}$ and kept for 3 h to obtain calcium oxide. This CaO (0.15 mol) was added to 250 cm^3 of ultra-pure water, stirred for 12 h with N_2 bubbling, and a suspension of $Ca(OH)_2$ in ultra-pure water was obtained. Nickel (II) acetate tetrahydrate ($Ni(CH_3COO)_2 \cdot 4H_2O$, Nacalai Tesque Inc.) or cobalt (II) acetate tetrahydrate ($Co(CH_3COO)_2 \cdot 4H_2O$, Nacalai Tesque Inc.) (0.017 mol) was added to the suspension, and then a phosphoric acid aqueous solution, which was prepared by mixing 0.10 mol of H_3PO_4 (85 mass%, Nacalai Tesque Inc.) with 250 cm^3 of ultra-pure water, was slowly added to this suspension at a rate of about 300 $cm^3 \cdot h^{-1}$. The sample notations and starting compositions are given in **Table 1**. The pH of the solution was controlled between 10.0 and 10.5 by addition of a NH_4OH solution at room temperature. N_2 bubbling continued during the stirring. HAp with no addition of metal ions (n-HAp) was also synthesized.

Table 1. Starting Compositions for Synthesis of HAp Containing Metal Ions

Notation	Composition (molar ratio)
n-HAp	Ca : P = 1.67 : 1.00
Ni-HAp	Ca : Ni : P = 1.50 : 0.17 : 1.00
Co-HAp	Ca : Co : P = 1.50 : 0.17 : 1.00

The sample powders were obtained by filtration and dried at 333 K. The powder was heated at various temperatures at a rate of 5 $K \cdot min^{-1}$.

2.2 Synthesis of OCP containing metal ions

Synthesis of the OCP containing Ni, Co or Mn (Ni-OCP, Co-OCP and Mn-OCP) was attempted as follows. We added 0.100 mol of H_3PO_4 (85 mass%) to 1 dm^3 of ultra-pure water, and stirred. $CaCO_3$ (0.133 mol) was then added to a H_3PO_4 aqueous solution at 333 K. After stirring for 3 h, 0.010 or 0.050 mol of $NiSO_4 \cdot 6H_2O$, $CoCl_2 \cdot 6H_2O$ or $MnCl_2 \cdot 4H_2O$ (Wako Pure Chemicals Ltd.) was added to the suspension. These sources of metal ions were selected because of their high solubility. The sample notation and starting compositions are given in **Table 2**. We expected that the added metal ions might be incorporated in the layer structure of OCP, and the Ca/P molar ratio was kept at 1.33 and metal ions were added. After addition of metal compounds, the pH was controlled above 9 by addition of a NH_4OH solution. After stirring for another 3 h, the sample powders were collected by filtration and dried at 333 K.

OCP with no addition of metal ions (n-OCP) was also prepared.

2.3 Characterization of samples

The samples were dissolved in a nitric acid solution and their compositions were examined by inductively coupled plasma atomic emission spectroscopy (ICP, Optima 2000DV, PerkinElmer Japan Co.). Their thermal properties were examined by thermogravimetry and differential thermal analysis (TG–DTA, TG–DTA2020S, MAC Science Co., Ltd.). Their structures were examined by powder X-ray diffraction (XRD, RINT2200V/PC–LR, Rigaku Co.).

2.4 Evaluation of catalytic activity

The catalytic activity for the decomposition of H_2O_2 was examined by adding the samples to a H_2O_2 solution. The sample powder 15 mg was added to 30 cm^3 of 0.05% H_2O_2 solution. The solution was shaken at 293 K. After 24 h, 10 cm^3 of supernatant was titrated with 0.02 $mol \cdot dm^3$ $KMnO_4$ solution to measure the residual H_2O_2 concentration. The decomposition rate was calculated from the residual H_2O_2 concentration. The decomposition rate was defined as 0% for the solution without addition of the samples.

3. Results

Table 3 shows the compositions of the obtained HAp containing metal ions without calcination, as determined by ICP. When Ni or Co ions were added during the preparation, Ni or Co was detected for the samples. The Ca/P molar ratio of n-HAp (1.54) was smaller than that of the stoichiometric HAp (1.67). This indicates that calcium-deficient HAp was obtained even for the n-HAp. The Ca/P molar ratios of n-HAp (1.54) and Co-HAp (1.57) were almost the same, whereas that of Ni-HAp was smaller (1.36).

Figure 1 shows XRD patterns of the n-HAp, Ni-HAp and Co-HAp before and after calcination at various temperatures. The TG–DTA results implied that significant changes occur at about 1073 K for Ni-HAp, and at about 1173 K for Co-HAp. From these results, the calcination temperature was selected. n-HAp, Ni-HAp and Co-HAp gave broad patterns ascribed to HAp (PDF#09–0432) when the calcination temperature was less than 873 K. When the n-HAp was calcined at 1073 K, the peaks became sharp. This indicates that the crystallinity increased by the calcination. New large peaks ascribed to $Ca_{19}Ni_2(PO_4)_{14}$ (PDF#49–1224) and a small peak ascribed to NiO (PDF#47–1049) were also detected in addition to sharp HAp peaks for the Ni-HAp calcined at 1073 K. The $Ca_{19}Ni_2(PO_4)_{14}$ is Ni-substituted tricalcium phosphate (TCP, $Ca_3(PO_4)_2$). As the peaks were observed at a higher diffraction angle than those of β-TCP (PDF#09–0169) and the peak positions almost matched those of $Ca_{19}Ni_2(PO_4)_{14}$, we assigned the observed peaks to $Ca_{19}Ni_2(PO_4)_{14}$. Therefore, the compositions of the phase which we assigned to $Ca_{19}Ni_2(PO_4)_{14}$ may not have the exact stoichiometric composition. New peaks ascribed to $Ca_{19}Co_2(PO_4)_{14}$ (PDF#49–1081) and Co_3O_4 (PDF#42–1467) were detected in addition to the sharp HAp peaks for Co-HAp calcined at 1173 K. $Ca_{19}Co_2(PO_4)_{14}$ is also Co-substituted TCP and the situation is same as for $Ca_{19}Ni_2(PO_4)_{14}$.

Figure 2 shows the comparison between the 002 XRD diffraction line of n-HAp and that of Ni-HAp or Co-HAp. The position of the 002 diffraction line of the Ni-HAp and n-HAp was almost the same before calcination, whereas for Ni-HAp the line shifted to a larger 2θ when Ni-HAp was calcined at 873 K; however, for n-HAp the position was not affected by the calcination. On the other hand, the position of the 002 diffraction line of the Co-HAp did not change after calcination at 873 K and was almost the same as that of

Table 2. Starting Compositions for Preparation of OCP Containing Metal Ions

Sample	Composition (molar ratio)
n-OCP	Ca : P = 1.33 : 1.00
Ni0.1-OCP	Ca : Ni : P = 1.33 : 0.10 : 1.00
Ni0.5-OCP	Ca : Ni : P = 1.33 : 0.50 : 1.00
Co0.1-OCP	Ca : Co : P = 1.33 : 0.10 : 1.00
Co0.5-OCP	Ca : Co : P = 1.33 : 0.50 : 1.00
Mn0.1-OCP	Ca : Mn : P = 1.33 : 0.10 : 1.00
Mn0.5-OCP	Ca : Mn : P = 1.33 : 0.50 : 1.00

Table 3. Compositions of HAp Containing Metal Ions Before Calcination

Sample	Composition (molar ratio)
n-HAp	Ca : P = 1.54 : 1.00
Ni-HAp	Ca : Ni : P = 1.36 : 0.10 : 1.00
Co-HAp	Ca : Co : P = 1.57 : 0.14 : 1.00

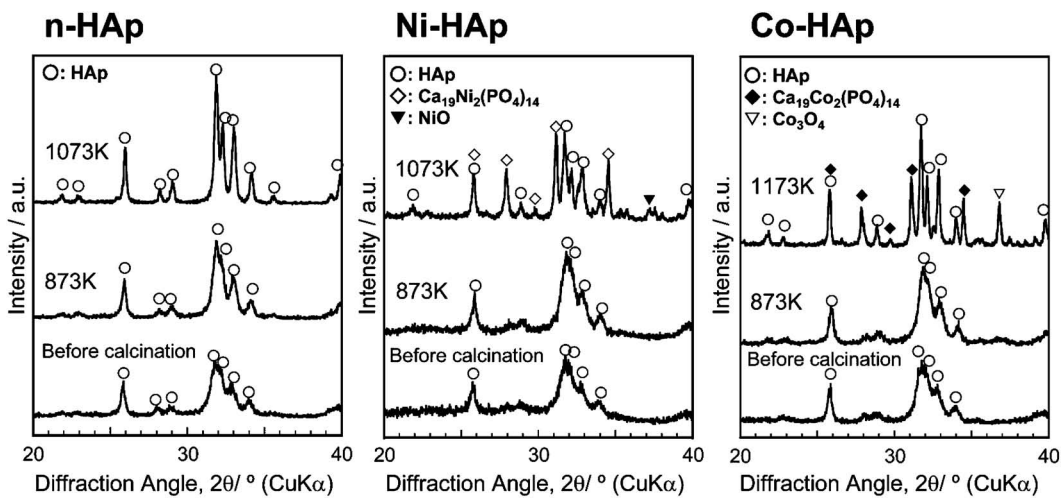

Fig. 1. XRD patterns of the n-HAp, Ni-HAp and Co-HAp before and after calcination at various temperatures.

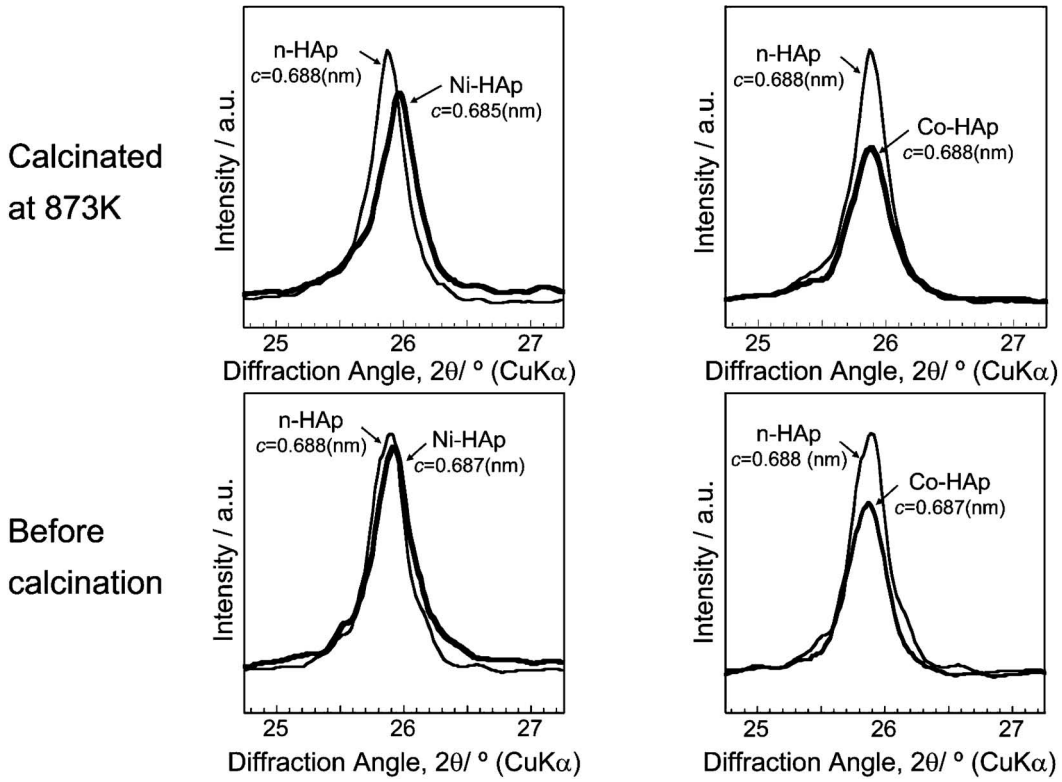

Fig. 2. Comparison between the 002 XRD diffraction peaks of n-HAp, Ni-HAp and Co-HAp. The lattice parameter of the *c*-axis is shown under each sample name.

Table 4. Compositions of OCPs Containing Metal Ions

Sample	Composition (molar ratio)
n-OCP	Ca : P = 1.36 : 1.00
Ni0.1-OCP	Ca : Ni : P = 1.43 : 0.11 : 1.00
Ni0.5-OCP	Ca : Ni : P = 1.40 : 0.32 : 1.00
Co0.1-OCP	Ca : Co : P = 1.36 : 0.09 : 1.00
Co0.5-OCP	Ca : Co : P = 1.28 : 0.46 : 1.00
Mn0.1-OCP	Ca : Mn : P = 1.33 : 0.11 : 1.00
Mn0.5-OCP	Ca : Mn : P = 1.28 : 0.42 : 1.00

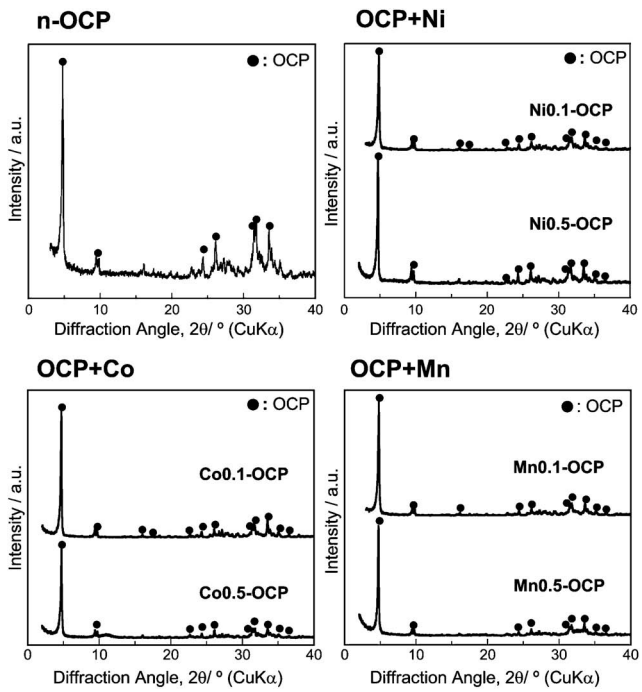

Fig. 3. XRD patterns of n-OCP and OCPs containing metal ions.

n-HAp. The lattice parameters of the *c*-axis, as determined from these results, are shown in these figures. The lattice parameter of the *c*-axis of Ni-HAp calcined at 873 K was smaller than for n-HAp and Co-HAp.

Table 4 shows the compositions of the OCP containing metal ions, as determined by ICP. When Ni, Co or Mn ions were added during the preparation, Ni, Co or Mn was detected in the samples. The incorporated amount of metal ions increased with increasing amount of added metal ions. The Ca/P molar ratio of n-OCP (1.36) was similar to that of the stoichiometric OCP (1.33). The Ca/P molar ratios of OCPs have not changed significantly by the addition of metal ions regardless of the kind and amount of the metal ions. **Figure 3** shows the XRD patterns of n-OCP and OCPs containing metal ions. All the peaks for all the samples were assigned to OCP (PDF#26–1056). No significant peak shift was detected for any samples.

Table 5 shows the decomposition rate of H_2O_2 for the samples. Regardless of calcination, n-HAp and Ni-HAp did not show catalytic activity for decomposition of H_2O_2, but Co-HAp showed high activity. The activity of Co-HAp was increased by calcination. In the case of OCP, n-OCP and OCPs containing Ni did not show catalytic activity for decomposition of H_2O_2, but OCPs containing Co or Mn showed high activity, which seemed to increase with increasing amount of added metal ions.

Table 5. Decomposition Rate of H_2O_2 for the Obtained Samples

Sample	Decomposition rate (%)
n-HAp (without calcination)	2
Ni-HAp (without calcination)	2
Ni-HAp (calcined at 600°C)	5
Co-HAp (without calcination)	72
Co-HAp (calcined at 600°C)	99
n-OCP	3
Ni0.1-OCP	1
Ni0.5-OCP	9
Co0.1-OCP	99
Co0.5-OCP	99
Mn0.1-OCP	91
Mn0.5-OCP	99

4. Discussion

When Ni or Co ions were added during the synthesis of HAp, HAps containing these ions were obtained. The XRD patterns of these samples showed that the single phase of HAp was obtained when calcination was conducted under 873 K. The lattice parameter of the *c*-axis of Ni-HAp calcined at 873 K was smaller than that of n-HAp. As the ion size of Ni^{2+} (0.069 nm) is smaller than that of Ca^{2+} (0.100 nm),[10] the decrease in the lattice parameter of *c*-axis is considered reasonable, assuming that Ni ions were incorporated in the crystal structure of HAp. However, the lattice parameter of *c*-axis of Ni-HAp without calcination was almost the same as that of n-HAp. This may indicate that Ni ions were not incorporated in the crystal structure of HAp before calcination and were incorporated by the calcination. There is a possibility that Ni ions exist as amorphous compounds, such as hydroxide, at low temperature. When the calcination temperature increased up to 1073 K, Ni ions were precipitated as $Ca_{19}Ni_2(PO_4)_{14}$ or NiO. On the other hand, no peak shift was detected for Co-HAp even after calcination at 873 K. The lattice parameter of the *c*-axis of Co-HAp is expected to be smaller than n-HAp because the ion size of Co^{2+} (0.065 nm)[10] is smaller than that of Ca^{2+}. The added Co^{2+} ions may exist as amorphous compounds, such as hydroxide. These considerations may also be supported from the results that the Ca/P molar ratios of n-HAp and Co-HAp were almost the same and that of Ni-HAp was smaller. If metal ions are incorporated in the crystal structure of HAp, they are expected to substitute the Ca^{2+} of HAp, and this decreases the Ca/P molar ratio, as in the case of Ni-HAp. However, it is still difficult to clearly describe the states of these ions, such as incorporation in the crystal structure, coprecipitation as amorphous compounds and adsorption on the HAp surface.

When Ni, Co or Mn ions were added during the synthesis of OCP, OCPs containing these ions were obtained. XRD patterns of the samples showed a single phase of OCP. However, it is difficult to deny the coexistence of HAp in these samples because the HAp peaks overlap with OCP peaks. We expected that the added ions might be incorporated in the layer structure of OCP, but it was not clear from the XRD patterns whether these ions were incorporated in their layer structure or crystal structures, as no significant peak shift was detected. Moreover, the Ca/P molar ratios of OCPs were not changed significantly by the addition of metal ions, regardless of the kind and amount of these ions. If metal ions are incorporated in the crystal structure of OCP, they are expected to substitute the Ca^{2+} of OCP, and thus decrease the Ca/P molar ratio. Although these results suggest that metal ions exist as amorphous compounds or adsorb on the surface, it is difficult to clarify the states of these ions even in the case of OCP with added metal ions.

The n-HAp and n-OCP did not show high catalytic activity for decomposition of H_2O_2, whereas HAp containing Co ions and OCP containing Co or Mn ions showed high catalytic activity for decomposition of H_2O_2. This indicates that the added Co or Mn ions in calcium phosphates are effective for the decomposition of H_2O_2. Although it has been reported that the zeolite-encapsulated Ni complex can decompose H_2O_2,[9] the HAp and OCP containing Ni ions did not show catalytic activity. This might be due to the state of the Ni ions. In the future study, the relationship between the element state and catalytic activity should be examined.

5. Conclusion

Calcium phosphates containing Co or Mn ions showed catalytic activity for the decomposition of H_2O_2. The addition of metal ions, such as Co and Mn ions, gave the catalytic activity to calcium phosphates. These calcium phosphates may be useful as an adsorbent that can be reused after H_2O_2 treatment.

References

1) LeGeros, R. Z., *Clin. Orthop. Relat. R.*, Vol. 395, pp. 81–98 (2002).
2) Kawasaki, T., *J. Chromatogr.*, Vol. 544, pp. 147–184 (1991).
3) Sujaridworakun, P., Koh, F., Fujiwara, T., Pongkao, D., Ahniyaz A. and Yoshimura, M., *Mater. Sci. Eng. C.*, Vol. 25, pp. 87–91 (2005).
4) Nakajima, A., Takakuwa, K., Kameshima, Y., Hagiwara, M., Sato, S., Yamamoto, Y., Yoshida, N., Watanabe, T. and Okada, K., *J. Photoch. Photobio. A*, Vol. 177, pp. 94–99 (2006).
5) Wakamura, M., Hashimoto, K. and Watanabe, T., *Langmuir*, Vol. 19, pp. 3428–3431 (2003).
6) Nishikawa, H., Komori, T., Kawachi, G. and Ioku, K., *J. Soc. Inorg. Mater. Japan*, Vol. 13, pp. 59–63 (2006).
7) Mathew M. and Takagi, S., "Octacalcium phosphate", Ed. by L. C. Chow and E. D. Eanes, Karger, Basel (2001) pp. 1–16.
8) Ohta, K., Monma, H., Takahashi, S. and Kobayashi, H., *J. Ceram. Soc. Japan*, Vol. 107, pp. 577–581 (1999).
9) Xavier, K. O., Chacko, J. and Yusuff, K. K. M., *Appl. Catal. A*, Vol. 258, pp. 251–259 (2004).
10) "Handbook of chemistry and physics, 80th ed.", Ed. by D. R. Lide, CRC Press, Boca Raton (1999).